

Project 3 Pitch: Gridiron Brief

AI-Powered Fantasy Football League Chat Summarization

Project Type: Large Language Model / Dialogue Summarization

Dataset: SAMSum

Proposed Model: Pre-Trained BERT Encoder-Decoder Architecture

Problem Statement and Proposed Solution

Problem Description

Fantasy football leagues generate frequent group-chat activity throughout the season. Members discuss trades, waiver claims, injuries, lineup decisions, league rules, and matchup results. In active leagues, these conversations can quickly grow into dozens or hundreds of messages.

Users who are away from the chat may return to a large backlog and struggle to identify what information actually matters. Important items such as trade proposals, waiver deadlines, injury updates, or commissioner announcements can become buried among jokes, reactions, and side conversations.

This project proposes **Gridiron Brief**, an automated dialogue summarization system that condenses lengthy fantasy football league conversations into short, readable summaries. The goal is to help users understand the key points of a conversation without reading every missed message.

For example:

Gridiron Brief: Mike proposed a trade involving a running back and wide receiver, but no agreement was reached. The league also discussed an injury affecting Sunday's lineups, and waiver claims are due tonight.

Business Impact and Solution Vision

Information overload can reduce user satisfaction and make it harder to re-enter active conversations. A user may either spend significant time reading the entire chat or skip messages and risk missing relevant information.

Gridiron Brief would reduce this burden by highlighting the most important actions, decisions, and discussion points. The feature could:

- Reduce conversation catch-up time
- Improve access to important league information
- Encourage users to re-engage with active chats

- Add an AI-powered feature that differentiates the messaging platform
- Create future opportunities for premium summaries or daily league digests

Because no proprietary Acme engagement data is available, baseline usage statistics will not be assumed. Instead, the proof of concept will focus on measurable model quality and define product metrics that could later be tested in a user study.

Success Criteria

The project will evaluate both technical performance and practical usefulness.

Measure	Target
ROUGE-1	≥ 35
ROUGE-2	≥ 12
ROUGE-L	≥ 32
Human usefulness rating	$\geq 4.0 / 5$
Major factual-error-free summaries	$\geq 85\%$
Future catch-up time reduction target	$\geq 50\%$
Important-information recall target	$\geq 85\%$

ROUGE will measure similarity between generated summaries and human-written reference summaries. Because word overlap alone does not guarantee a useful summary, a sample of outputs will also be reviewed for relevance, factual accuracy, fluency, conciseness, and completeness.

Problem-Solving Process

Process Framework

The project will follow a seven-step NLP workflow:

1. Research and Business Understanding

Review transformer architecture, BERT, encoder-decoder models, dialogue summarization, ROUGE evaluation, and the SAMSum dataset. Define what information a useful fantasy football summary should preserve.

2. Dataset Exploration

Load SAMSum and examine dialogue length, summary length, number of speakers, conversational turns, and sample dialogue-summary pairs. The existing training, validation, and test splits will be retained.

3. Data Preparation

Tokenize dialogues and summaries using a pre-trained tokenizer. Configure padding, truncation, attention masks, labels, and batched data loading.

4. Model Implementation

Build a BERT-based encoder-decoder model using Hugging Face's `EncoderDecoderModel`. The encoder will represent the conversation context, while the decoder will generate the summary.

5. Training and Optimization

Fine-tune the model on SAMSum using validation-loss monitoring, checkpointing, early stopping, and memory-conscious training settings.

6. Evaluation and Error Analysis

Evaluate unseen test conversations using ROUGE-1, ROUGE-2, and ROUGE-L. Review model outputs for factual errors, speaker attribution mistakes, omitted information, repetition, and hallucination.

7. Fantasy Football Demonstration

Test the trained model on a small set of realistic fantasy football conversations involving trades, waivers, injuries, league votes, and commissioner announcements.

Conceptual Representation

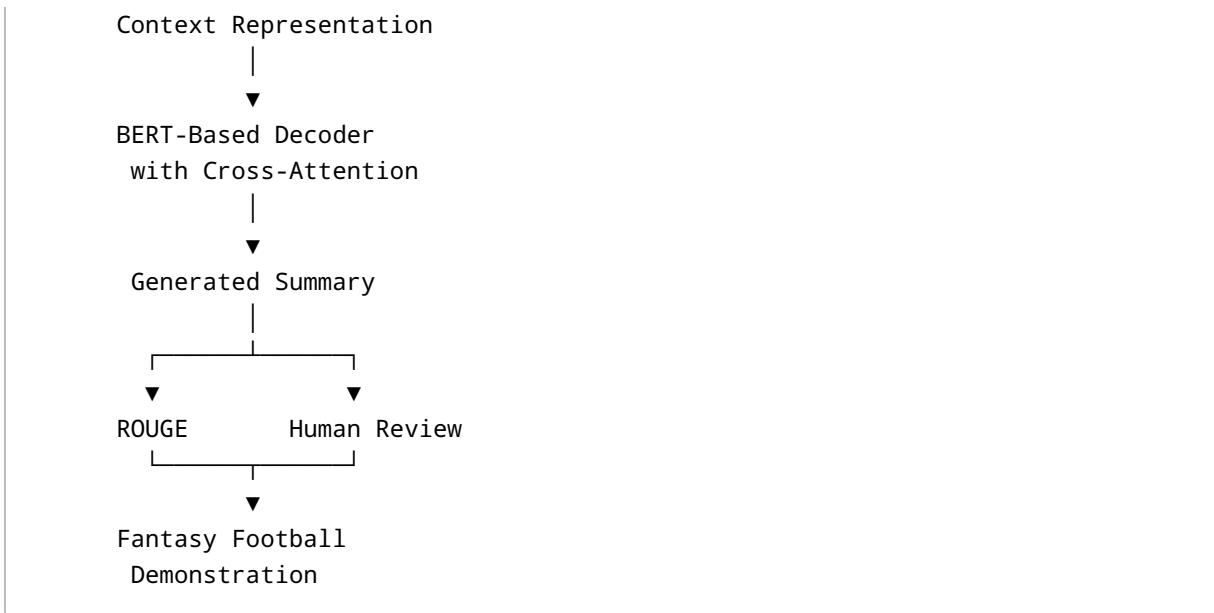
SAMSum Dialogue + Human Summary

▼
Data Exploration
and Preparation

▼
Tokenization

▼
Pre-Trained BERT Encoder

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Methodology Justification

BERT Encoder-Decoder Architecture

Dialogue summarization requires both language understanding and text generation. The encoder must understand how messages, speakers, and topics relate across a conversation, while the decoder must convert that context into a concise natural-language summary.

A BERT-based encoder-decoder architecture directly supports this sequence-to-sequence task and aligns with the project requirement to use a pre-trained BERT model.

Fine-Tuning Pre-Trained Models

Fine-tuning is more practical than training a transformer from scratch. A pre-trained model already contains useful language representations, allowing the project to focus on adapting the model to dialogue summarization rather than learning general language structure from the beginning.

This reduces:

- Compute requirements
- Training time
- Data requirements
- Development risk

Evaluation Strategy

ROUGE provides a standardized way to compare generated summaries with human references, but it cannot fully measure factual accuracy or usefulness. Therefore, quantitative evaluation will be paired with qualitative review.

This combination is especially important for dialogue, where a summary may use different wording than the reference or incorrectly assign an action to the wrong speaker.

Timeline and Scope

Dates	Phase	Key Tasks
Aug. 20–22	Research & Pitch	Review SAMSum, BERT architecture, ROUGE, business problem, and finalize pitch
Aug. 23–24	Data Preparation	Load SAMSum, perform EDA, tokenize text, configure data pipeline
Aug. 25–26	Model Implementation	Build BERT encoder-decoder model and validate the pipeline on a small subset
Aug. 27–29	Training	Fine-tune model, monitor validation loss, save checkpoints, adjust resource settings
Aug. 30–31	Evaluation	Calculate ROUGE scores and perform qualitative error analysis
Sept. 1–2	MVP & Demo	Create fantasy football test conversations and prepare critique submission
Sept. 3–5	Final Refinement	Incorporate feedback, finalize notebook, report, and presentation

Iteration Points

Three checkpoints will guide model refinement:

1. **After preprocessing:** confirm tokenization and sequence lengths preserve enough conversational information.
 2. **After pilot training:** verify that the model trains, saves, reloads, and generates summaries before scaling up.
 3. **After evaluation:** make one controlled adjustment to training or generation if results fall below expectations.
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Risk Management

Compute Limitations

Transformer training may exceed available memory or take longer than expected.

Mitigation: Begin with a smaller subset, reduce batch size, use gradient accumulation, limit sequence lengths, and save checkpoints.

Model Configuration

Encoder-decoder models require correct tokenization, special-token settings, and generation parameters.

Mitigation: Validate the complete pipeline on a small batch before full training.

Weak or Inaccurate Summaries

The model may omit important information, repeat text, hallucinate details, or confuse speakers.

Mitigation: Evaluate with both ROUGE and human review, then make targeted changes to training or decoding settings.

Domain Shift

SAMSum contains general conversations rather than fantasy-football-specific language.

Mitigation: SAMSum will remain the formal training and evaluation dataset. Fantasy football conversations will be used only as a final domain-transfer demonstration, and any limitations will be documented.

Scope

The project will remain focused on one central task:

Can a pre-trained BERT encoder-decoder model convert a lengthy group conversation into a concise, useful summary?

Fantasy football provides the business context, but the project will not include player projections, trade recommendations, live injury feeds, or predictive sports analytics.

This limited scope keeps the project achievable while still producing a distinctive application of dialogue summarization.

Final Deliverables

The completed project will include:

- Jupyter Notebook with preprocessing, modeling, training, evaluation, and inference
 - ROUGE performance results
 - Qualitative error analysis
 - Sample generated summaries
 - Fantasy football demonstration conversations
 - 3–5 page written report
 - 5-minute showcase presentation
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Conclusion

Gridiron Brief applies dialogue summarization to a familiar problem in fantasy football: important information becoming buried in active league chats. The project will fine-tune a pre-trained BERT encoder-decoder model on the SAMSum dataset and evaluate its ability to generate concise, accurate summaries of unseen conversations.

By combining quantitative ROUGE evaluation with qualitative review and a focused fantasy football demonstration, the project will provide a practical proof of concept for using NLP to reduce information overload and improve the usability of group messaging.

References

Gliwa, B., Mochol, I., Biesek, M., & Wawer, A. (2019). *SAMSum Corpus: A Human-annotated Dialogue Dataset for Abstractive Summarization*. Association for Computational Linguistics.

Lin, C.-Y. (2004). *ROUGE: A Package for Automatic Evaluation of Summaries*. Association for Computational Linguistics.

Hugging Face. *EncoderDecoderModel* — *Transformers Documentation*.

Hugging Face. *SAMSum Dataset*.